

Raj Patel

Cyber & AI Risk | Security Research

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EDUCATION

Georgia Institute of Technology

Master of Science, **Computer Engineering**

Bachelor of Science, **Computer Engineering**

Atlanta, GA

December 2026

December 2024

TECHNICAL SKILLS

- **Programming:** Python, C, C++, x86/x64 Assembly, Java, JavaScript, PowerShell, Bash, SQL, KQL, R
- **Tools:** RecordedFuture, ThreatConnect, DomainTools, IDA Pro, GitHub, x64dbg, Ghidra, SIEM (Splunk, ELK), IDS (Suricata, Snort), Wireshark, YARA, Kubernetes, Docker, VMware, n8n, Tableau, Linux, Claude Code
- **AI/ML:** LLM agent design, prompt engineering, RAG, MCP, PyTorch, scikit-learn, pandas, NumPy, model red teaming
- **Security Tooling:** IDA Pro, Ghidra, x64dbg, YARA, Wireshark, Suricata, Snort, Splunk, Elastic (ELK), Burp Suite, Recorded Future, ThreatConnect, DomainTools
- **Cloud/Infra:** AWS (IAM, Cognito, API Gateway), Docker, Kubernetes, VMware, Git/GitHub Actions, Linux
- **Frameworks:** MITRE ATT&CK, MITRE ATLAS, NIST CSF, OWASP Top 10, OWASP LLM Top 10, Cyber Kill Chain, Diamond Model
- **Certifications:** CompTIA Security+ (September 2026)

EXPERIENCE

Georgia Tech - Center for Cyber Operations and Unconventional Sensing (COEUS)

May 2026 – Present

Graduate Research Assistant

Atlanta

- Architected an Agentic AI framework for automated payload analysis and vulnerability triage, obtaining an accuracy of 92% and generating dynamic execution scripts that improved execution trigger rates on held-out samples by 67%.
- Validated SOTA binary decomposition frameworks by reproducing 3 published research artifacts and executing end-to-end evaluations, confirming model fidelity across a custom software supply chain dataset.

Fortune 100 M&E Company - Practicum (client engagement) - name withheld per NDA

January 2026 – May 2026

Cyber Threat Intelligence Analyst

Atlanta

- Built a comprehensive threat intelligence inventory of 14 incidents by mapping 3 years of threat landscape evolution and TTPs to the MITRE ATT&CK framework, using Recorded Future and OSINT sources.
- Improved incident research speed by 2 hours by prompt-engineering Recorded Future AI Insights to query incidents and surface relevant reporting, iteratively refining prompts based on evaluation results.

Pramukh Transport Ltd.

August 2022 – January 2025

Cyber Risk Analyst, Part-time

Kenya

- Conducted a red team tabletop exercise and threat modeling across 3 threat objectives, analyzing ransomware intelligence to identify emerging risks and mitigate critical vulnerabilities in transport infrastructure including exposed RDP and weak credential management.

PROJECTS

Malware Reverse Engineering

- Reverse engineered 5 malware families using IDA Pro, x86 Assembly, and Ollydbg to extract C2 infrastructure, analyze Windows API abuse patterns, and document sandbox evasion techniques in technical reports.
- Captured and analyzed malicious network traffic with Wireshark to identify C2 communication protocols, data exfiltration methods, and command structures for threat intelligence reporting.

ML-Based Network Intrusion Detection System

- Developed an ML-based defense system that removed hidden backdoor attacks using machine unlearning, leveraging scikit-learn, Pandas and NumPy for model training and evaluation.
- Engineered 3 SOC-ready Suricata signatures to detect SSH brute-force attacks and validated detection accuracy against CIC-IDS2017 dataset (50 GB of PCAP data), achieving 90% detection rate during testing ensuring high-confidence alerts suitable for rapid triage and escalation of flagged malicious activity.

Secure Cloud Web Application Development

- Enhanced the security of a cloud-native web application by mitigating OWASP Top 10 broken authentication risks through AWS IAM least-privilege RBAC, Cognito authentication, and API Gateway rate limiting.